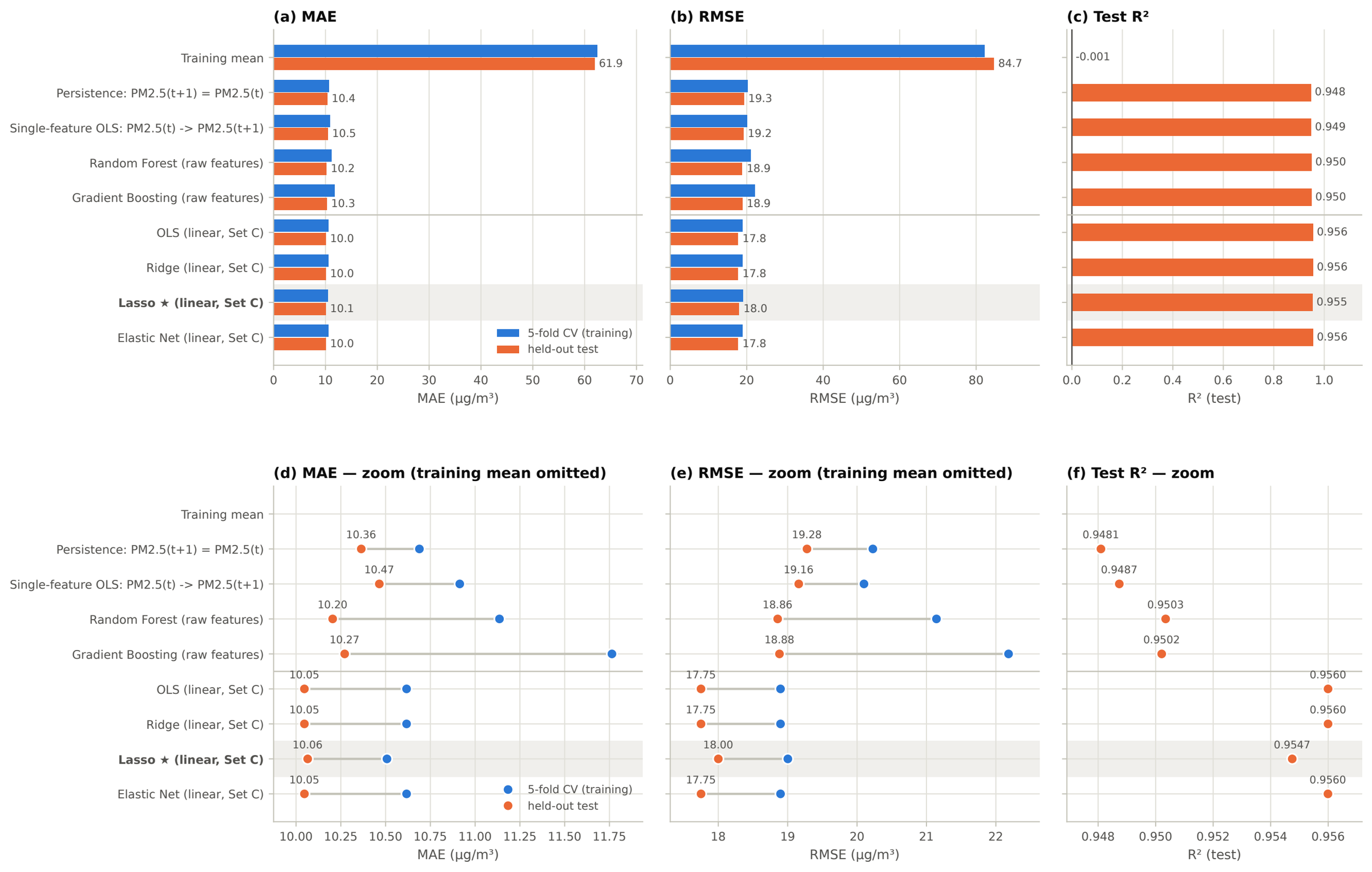
(a) MAE
Training mean
Persistence: PM2.5(t+1) = PM2.5(t)
Single-feature OLS: PM2.5(t) -> PM2.5(t+1)
Random Forest (raw features)
Gradient Boosting (raw features)
OLS (linear, Set C)
Ridge (linear, Set C)
Lasso ★ (linear, Set C)
Elastic Net (linear, Set C)
61.9
10.4
10.5
10.2
10.3
10.0
10.0
10.1
10.0
5-fold CV (training)
held-out test
0 10 20 30 40 50 60 70
MAE (µg/m³)
(b) RMSE
84.7
19.3
19.2
18.9
18.9
17.8
17.8
18.0
17.8
0 20 40 60 80
RMSE (µg/m³)
(c) Test R²
-0.001
0.948
0.949
0.950
0.950
0.956
0.956
0.955
0.956
0.0 0.2 0.4 0.6 0.8 1.0
R² (test)
(d) MAE — zoom (training mean omitted)
10.36
10.47
10.20
10.27
10.05
10.05
10.06
10.05
10.00 10.25 10.50 10.75 11.00 11.25 11.50 11.75
MAE (µg/m³)
(e) RMSE — zoom (training mean omitted)
19.28
19.16
18.86
18.88
17.75
17.75
18.00
17.75
18 19 20 21 22
RMSE (µg/m³)
(f) Test R² — zoom
0.9481
0.9487
0.9503
0.9502
0.9560
0.9560
0.9547
0.9560
0.948 0.950 0.952 0.954 0.956
R² (test)